Supplementary Materials

Please note that additional detailed Supplementary Materials (including full code, case studies, and results data) have been hosted in this separate, dedicated Supplementary Materials repository to facilitate access:

`https://anonymous.4open.science/r/CRAVE-D983`

Section S1: Exemplar Selection for Qualitative Analysis

We conduct a qualitative case analysis to examine the framework's capacity to mitigate the homogenized positivity bias commonly observed in LLM-generated narratives [1, 2]. Among the evaluated emotional arcs, we specifically focus on the "Riches-to-Rags" curve to directly challenge this bias, adopting a rule-based Hierarchical Consensus Protocol to objectively identify an exemplar from the routes generated under this specific input. Specifically, we first filter routes that satisfy an Overall Quality (OV) threshold of ≥ 4.0. Among these candidates, we identify trajectories that achieve consensus among three independent LLM evaluators on the route with the highest Empathy (EM), prioritizing unanimous (3/3) agreement and considering majority (2/3) consensus only when no unanimous agreement is present. If multiple routes receive the same level of consensus as the top EM route, the tie is resolved by applying the same consensus criterion to the OV score, selecting the route with the higher OV. The resulting route is reported as an illustrative exemplar for qualitative analysis. To visualize the emotional trajectory, we overlay a 10% sliding-window simple moving average onto the raw sentence-level scatter points, preserving the granular distribution of the original data while effectively extracting macro-narrative trends without introducing polynomial distortion.

Section S2: Implementation Details

The software environment was built on Python 3.12. Narrative generation for all evaluated frameworks utilized *gpt-5-mini-2025-08-07* accessed via API. To ensure absolute consistency in hardware acceleration and avoid inference bottlenecks, all subsequent local computational tasks were executed on a single dedicated workstation equipped with an NVIDIA A100 GPU (80GB VRAM) and 384GB RAM. These tasks included: (1) the LLM-as-a-Judge textual evaluation using three open-weight models (*gemma3:12b-it-fp16*, *mistral:7b-instruct-v0.3-fp16*, *qwen3:14b-fp16*); and (2) the DeBERTa-based continuous affective analysis. All generated scripts were compiled and syntactically validated using the Ren'Py 8.5.2 engine. For reproducibility, a supplementary file, all source code, generation prompts, and full evaluation data are available in the Supplementary Material GitHub repository[1].

Section S3: Statistical Demographics of the Generated Corpus

To ensure that the experimental comparisons between the generation variants are grounded in a consistent data scale, we report the baseline descriptive statistics for the evaluation corpus of 72 visual novels. The generation methods evaluated in our study are defined by their distinct trajectory guidance—Rise and Fall (RAF) versus Arousal and Valence Level (AVL) guidance, the latter implemented as a discrete 5×5 matrix—and evaluated both with and without the hierarchical Critic module.

Table S1 summarizes the structural demographics of the generated stories, disaggregated by these four configurations (`RAF_NoCrit`, `RAF_Critic`, `AVL_NoCrit`, `AVL_Critic`).

These metrics demonstrate that despite the distinct architectural trajectories dictated by the different affective guidance configurations, the generation pipeline reliably produces structurally complex, novel-length interactive narratives of comparable scale and interactive density. The generation volume,

[1] `https://github.com/Yi-Xia-2010/CRAVE`

Table S1: Descriptive statistics of the generated visual novel corpus (N=72), detailing average word counts, unique characters, decision point frequencies, and total narrative branches across the four generation variants.

Metric	`RAF_NoCrit`	`RAF_Critic`	`AVL_NoCrit`	`AVL_Critic`
Words per VN	10,144	10,840	11,262	12,458
Words per Chapter	601.1	649.6	588.2	631.6
Unique Characters per VN	22.1	19.7	22.7	22.4
Decision Points per VN	81.2	82.8	97.4	96.4
Decisions per Chapter	4.83	5.01	5.14	5.10
Narrative Routes per VN	7.22	6.17	7.50	7.39

structural complexity, and number of playable routes remain stable across all variants. These consistent metrics establish a stable structural baseline for our comparative evaluation

Section S4: Detailed Statistical Tables

This section provides the comprehensive statistical results of the 2×2 factorial evaluation.

S4.1 LLM-as-a-Judge Results

Table S2 summarizes the LLM-as-a-Judge quantitative results. Tables S3 to S5 detail the main effects across the **AllMean Pool**, while Tables S6 to S8 detail the main effects within the **Top-3 Pool**. For extended statistical results, including interaction effects, simple effects, and raw score distributions, please refer to the provided repository.

Table S2: Quantitative Results: Generative Stability (AllMean) vs. Creative Upper-Bound (Top-3)

Criteria	**Judge**	**Pool**	**RAF**		**AVL**		**Main Effect** (p)	
			w/o Critic	w/ Critic	w/o Critic	w/ Critic	Guidance (A)	Critic (B)
Relevance (RE)	Gemma-3	AllMean	4.95	4.91	4.92	4.95	0.749	0.916
		Top-3	5.00	4.98	4.98	5.00	1.000	1.000
	Mistral-7B	AllMean	4.45	4.52	4.43	4.52	0.951	0.068
		Top-3	4.71	4.64	4.81	4.87	**0.008**	0.833
	Qwen-3	AllMean	5.00	5.00	5.00	5.00	1.000	1.000
		Top-3	5.00	5.00	5.00	5.00	1.000	1.000
Coherence (CH)	Gemma-3	AllMean	4.95	4.94	4.84	4.87	**0.001**†	0.895
		Top-3	4.96	4.93	4.87	4.87	0.061	0.641
	Mistral-7B	AllMean	4.61	4.50	4.72	4.79	0.981	0.421
		Top-3	4.98	4.72	4.56	4.78	**0.013**†	0.289
	Qwen-3	AllMean	5.00	5.00	5.00	5.00	1.000	1.000
		Top-3	5.00	5.00	5.00	5.00	1.000	1.000
Empathy (EM)	Gemma-3	AllMean	4.10	4.18	4.35	4.25	**<0.001**	0.803
		Top-3	4.20	4.33	4.59	4.43	**<0.001**	0.782
	Mistral-7B	AllMean	4.04	4.02	4.11	4.06	**0.001**	**0.039***
		Top-3	4.10	4.05	4.24	4.13	**0.005**	**0.018***
	Qwen-3	AllMean	5.00	5.00	5.00	5.00	1.000	1.000
		Top-3	5.00	5.00	5.00	5.00	1.000	1.000
Surprise (SU)	Gemma-3	AllMean	3.00	3.02	3.05	3.05	**0.021**	0.581
		Top-3	3.02	3.07	3.15	3.13	**0.019**	0.641
	Mistral-7B	AllMean	3.32	3.34	3.35	3.35	0.524	0.906
		Top-3	3.66	3.56	3.78	3.73	**0.024**	0.346
	Qwen-3	AllMean	2.99	2.98	2.97	2.96	0.597	0.864
		Top-3	3.15	3.09	3.16	3.11	0.725	0.266
Engagement (EG)	Gemma-3	AllMean	4.41	4.47	4.57	4.49	**0.034**	0.803
		Top-3	4.69	4.70	4.85	4.80	**0.026**	0.752
	Mistral-7B	AllMean	4.17	4.20	4.23	4.25	**0.043**	0.613
		Top-3	4.39	4.39	4.54	4.54	**0.020**	0.970
	Qwen-3	AllMean	5.00	5.00	4.96	4.98	**0.007**†	0.553
		Top-3	5.00	5.00	4.98	5.00	0.322	0.322
Complexity (CX)	Gemma-3	AllMean	3.97	3.96	4.01	3.99	0.087	0.543
		Top-3	4.04	4.02	4.04	4.06	0.474	1.000
	Mistral-7B	AllMean	4.02	4.01	4.03	4.01	0.443	0.421
		Top-3	4.07	4.02	4.09	4.07	0.210	0.210
	Qwen-3	AllMean	4.05	4.08	3.98	3.99	**0.007**†	0.507
		Top-3	4.13	4.19	4.10	4.11	0.290	0.534
Overall (OV)	Gemma-3	AllMean	4.23	4.24	4.29	4.27	**0.010**	0.953
		Top-3	4.32	4.34	4.41	4.38	**0.001**	0.703
	Mistral-7B	AllMean	4.15	4.17	4.18	4.19	0.200	0.357
		Top-3	4.32	4.27	4.39	4.38	**0.002**	0.478
	Qwen-3	AllMean	4.51	4.51	4.48	4.49	0.055	0.582
		Top-3	4.55	4.55	4.54	4.54	0.798	0.519

Note: Forest green with **bold** font indicates significant main effects ($p < 0.05$). † indicates RAF > AVL. * indicates w/o Critic > w/ Critic. Evaluations from Qwen-3 exhibit a ceiling effect across several criteria and occasional directional divergence from the consensus of the other judge models in specific criteria (EG and CX).

Table S3: Detailed Main Effects of the 2×2 Factorial Evaluation (**AllMean Pool**, **Mistral-7B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p-values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	**Analyzed Effect**	**Diff.** **[95% CI]**	**Cohen's d**	**Statistic (U)**	**p-value**
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.004 [-0.081, 0.088]	0.008	32385.5	0.951
	Factor B (Critic: w/o vs w/)	-0.079 [-0.163, 0.005]	-0.164	29623.5	0.068
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	-0.194 [-0.409, 0.022]	-0.160	32314.5	0.981
	Factor B (Critic: w/o vs w/)	0.008 [-0.204, 0.221]	0.007	31641.5	0.421
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.056 [-0.091, -0.022]	-0.280	29561.5	**0.001**
	Factor B (Critic: w/o vs w/)	0.034 [-0.001, 0.069]	0.168	34097.5	**0.039**
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.020 [-0.100, 0.061]	-0.042	31403.0	0.524
	Factor B (Critic: w/o vs w/)	-0.012 [-0.092, 0.069]	-0.026	32165.0	0.906
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.060 [-0.128, 0.009]	-0.152	29796.0	**0.043**
	Factor B (Critic: w/o vs w/)	-0.026 [-0.095, 0.042]	-0.066	31705.0	0.613
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.010 [-0.042, 0.022]	-0.055	31783.0	0.443
	Factor B (Critic: w/o vs w/)	0.010 [-0.021, 0.042]	0.057	32866.5	0.421
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.027 [-0.064, 0.011]	-0.122	30280.0	0.200
	Factor B (Critic: w/o vs w/)	-0.015 [-0.053, 0.023]	-0.068	30881.0	0.357

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S4: Detailed Main Effects of the 2×2 Factorial Evaluation (**AllMean Pool**, **Gemma3-12B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p-values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	**Analyzed Effect**	**Diff. [95% CI]**	**Cohen's d**	**Statistic (U)**	**p-value**
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	-0.007 [-0.051, 0.037]	-0.028	32064.5	0.749
	Factor B (Critic: w/o vs w/)	-0.002 [-0.046, 0.041]	-0.009	32254.0	0.916
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.088 [0.037, 0.139]	0.295	35131.0	**0.001**
	Factor B (Critic: w/o vs w/)	-0.004 [-0.056, 0.049]	-0.012	32216.0	0.895
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.162 [-0.232, -0.091]	-0.396	27076.0	**0.001**
	Factor B (Critic: w/o vs w/)	0.009 [-0.063, 0.082]	0.022	32627.5	0.803
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.044 [-0.081, -0.007]	-0.205	30895.0	**0.022**
	Factor B (Critic: w/o vs w/)	-0.011 [-0.048, 0.027]	-0.049	31993.5	0.581
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.094 [-0.181, -0.007]	-0.189	29253.0	**0.034**
	Factor B (Critic: w/o vs w/)	0.011 [-0.076, 0.098]	0.022	32687.5	0.804
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.037 [-0.081, 0.006]	-0.152	31119.5	0.087
	Factor B (Critic: w/o vs w/)	0.013 [-0.030, 0.056]	0.054	32748.5	0.543
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.043 [-0.071, -0.014]	-0.263	28301.5	**0.010**
	Factor B (Critic: w/o vs w/)	0.003 [-0.026, 0.032]	0.018	32423.0	0.953

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S5: Detailed Main Effects of the 2×2 Factorial Evaluation (**AllMean Pool**, **Qwen3-14B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p-values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	**Analyzed Effect**	**Diff. [95% CI]**	**Cohen's d**	**Statistic (U)**	**p-value**
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	32294.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	32330.0	1.000
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	32294.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	32330.0	1.000
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	32294.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	32330.0	1.000
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	0.020 [-0.047, 0.086]	0.052	32833.5	0.597
	Factor B (Critic: w/o vs w/)	0.006 [-0.060, 0.072]	0.016	32505.0	0.864
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	0.030 [0.009, 0.050]	0.241	33258.0	**0.007**
	Factor B (Critic: w/o vs w/)	-0.007 [-0.028, 0.015]	-0.053	32117.5	0.553
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	0.082 [0.022, 0.142]	0.237	34848.5	**0.007**
	Factor B (Critic: w/o vs w/)	-0.019 [-0.079, 0.042]	-0.053	31701.0	0.507
Overall (OV)	Factor A (Guidance: RAF vs AVL)	0.022 [0.006, 0.038]	0.235	34584.0	0.055
	Factor B (Critic: w/o vs w/)	-0.003 [-0.020, 0.013]	-0.034	32989.5	0.582

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S6: Detailed Main Effects of the 2×2 Factorial Evaluation (**Top-3 Pool**, **Mistral-7B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p-values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p-value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	-0.167 [-0.269, -0.065]	-0.438	4846	**0.008**
	Factor B (Critic: w/o vs w/)	0.009 [-0.095, 0.114]	0.024	5753	0.833
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.185 [-0.084, 0.454]	0.185	6419	**0.013**
	Factor B (Critic: w/o vs w/)	0.019 [-0.252, 0.289]	0.018	5581.5	0.290
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.111 [-0.186, -0.036]	-0.398	4940	**0.005**
	Factor B (Critic: w/o vs w/)	0.083 [0.008, 0.159]	0.296	6581	**0.018**
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.147 [-0.264, -0.029]	-0.335	4959	**0.024**
	Factor B (Critic: w/o vs w/)	0.073 [-0.046, 0.191]	0.164	6198	0.346
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.148 [-0.275, -0.022]	-0.314	4872	**0.020**
	Factor B (Critic: w/o vs w/)	0.000 [-0.128, 0.128]	0.000	5848	0.970
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.037 [-0.098, 0.024]	-0.163	5518.0	0.211
	Factor B (Critic: w/o vs w/)	0.037 [-0.024, 0.098]	0.163	6146.0	0.211
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.092 [-0.144, -0.040]	-0.477	4471	**0.002**
	Factor B (Critic: w/o vs w/)	0.030 [-0.023, 0.083]	0.153	6148.5	0.478

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S7: Detailed Main Effects of the 2×2 Factorial Evaluation (**Top-3 Pool**, **Gemma3-12B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p-values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	**Analyzed Effect**	**Diff.** **[95% CI]**	**Cohen's d**	**Statistic (U)**	**p-value**
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.000 [-0.026, 0.026]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [-0.026, 0.026]	0.000	5832.0	1.000
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.074 [-0.004, 0.152]	0.257	6264.0	0.061
	Factor B (Critic: w/o vs w/)	0.019 [-0.060, 0.097]	0.064	5940.0	0.641
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.241 [-0.368, -0.113]	-0.507	4428.0	**<0.001**
	Factor B (Critic: w/o vs w/)	0.019 [-0.113, 0.150]	0.038	5940.0	0.782
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.093 [-0.170, -0.015]	-0.322	5292.0	**0.019**
	Factor B (Critic: w/o vs w/)	-0.019 [-0.097, 0.060]	-0.064	5724.0	0.641
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.130 [-0.244, -0.016]	-0.305	5076.0	**0.026**
	Factor B (Critic: w/o vs w/)	0.019 [-0.097, 0.134]	0.043	5940.0	0.752
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.019 [-0.069, 0.032]	-0.098	5724.0	0.474
	Factor B (Critic: w/o vs w/)	0.000 [-0.051, 0.051]	0.000	5832.0	1.000
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.068 [-0.105, -0.031]	-0.489	4469.5	**0.001**
	Factor B (Critic: w/o vs w/)	0.006 [-0.032, 0.045]	0.043	5991.0	0.703

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S8: Detailed Main Effects of the 2×2 Factorial Evaluation (**Top-3 Pool**, **Qwen3-14B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p-values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	**Analyzed Effect**	**Diff. [95% CI]**	**Cohen's d**	**Statistic (U)**	**p-value**
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.015 [-0.112, 0.081]	-0.043	5732.0	0.725
	Factor B (Critic: w/o vs w/)	0.053 [-0.044, 0.149]	0.146	6147.0	0.266
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	0.009 [-0.009, 0.028]	0.136	5886.0	0.322
	Factor B (Critic: w/o vs w/)	-0.009 [-0.028, 0.009]	-0.136	5778.0	0.322
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	0.049 [-0.056, 0.155]	0.125	6154.0	0.290
	Factor B (Critic: w/o vs w/)	-0.031 [-0.137, 0.075]	-0.078	5642.5	0.534
Overall (OV)	Factor A (Guidance: RAF vs AVL)	0.007 [-0.015, 0.029]	0.087	5922.0	0.798
	Factor B (Critic: w/o vs w/)	0.002 [-0.020, 0.024]	0.025	6058.5	0.519

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S9: Detailed Statistical Tests on Affective Salience and Intensity. Significance levels: ns ($p \geq 0.05$), * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

Metric	Comparison	Mean A	Mean B	Diff. [95% CI]	Cohen's d	Test Used	p-value
Valence Affective Salience (AS)	RAF_w/o_Critic vs RAF_w/_Critic	0.100	0.096	0.004 [0.000, 0.008]	0.269	Mann-Whitney U	**0.040** (*)
	AVL_w/o_Critic vs AVL_w/_Critic	0.107	0.104	0.003 [-0.001, 0.007]	0.211	Welch's t	0.085 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	0.100	0.107	-0.007 [-0.011, -0.003]	-0.428	Mann-Whitney U	**0.003** (**)
	RAF_w/_Critic vs AVL_w/_Critic	0.096	0.104	-0.008 [-0.011, -0.004]	-0.497	Welch's t	< **0.001** (***)
Arousal Affective Salience (AS)	RAF_w/o_Critic vs RAF_w/_Critic	0.235	0.236	-0.002 [-0.004, 0.001]	-0.148	Welch's t	0.258 (ns)
	AVL_w/o_Critic vs AVL_w/_Critic	0.234	0.235	-0.000 [-0.003, 0.002]	-0.041	Mann-Whitney U	0.895 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	0.235	0.234	0.001 [-0.002, 0.003]	0.071	Mann-Whitney U	0.752 (ns)
	RAF_w/_Critic vs AVL_w/_Critic	0.236	0.235	0.002 [-0.001, 0.005]	0.167	Welch's t	0.197 (ns)
Valence High-Salience Proportion (%)	RAF_w/o_Critic vs RAF_w/_Critic	23.32	21.85	1.47 [0.07, 2.87]	0.268	Mann-Whitney U	0.057 (ns)
	AVL_w/o_Critic vs AVL_w/_Critic	25.81	24.41	1.39 [-0.18, 2.97]	0.213	Mann-Whitney U	0.093 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	23.32	25.81	-2.48 [-3.95, -1.02]	-0.408	Mann-Whitney U	**0.004** (**)
	RAF_w/_Critic vs AVL_w/_Critic	21.85	24.41	-2.56 [-4.07, -1.05]	-0.423	Welch's t	< **0.001** (***)
Arousal High-Salience Proportion (%)	RAF_w/o_Critic vs RAF_w/_Critic	82.65	82.80	-0.16 [-1.11, 0.80]	-0.042	Welch's t	0.751 (ns)
	AVL_w/o_Critic vs AVL_w/_Critic	82.16	82.55	-0.39 [-1.29, 0.52]	-0.103	Mann-Whitney U	0.417 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	82.65	82.16	0.49 [-0.35, 1.33]	0.141	Mann-Whitney U	0.432 (ns)
	RAF_w/_Critic vs AVL_w/_Critic	82.80	82.55	0.26 [-0.76, 1.27]	0.064	Mann-Whitney U	0.754 (ns)

*Note:*The statistical test (Welch's t or Mann-Whitney U) was selected based on the outcome of the Shapiro-Wilk test for normality on the respective distributions. [95% CI] represents the 95% confidence interval of the mean difference.

S4.2 Detailed Affective Salience and Intensity Statistics

Table S9 presents the statistical testing on Affective Salience (AS) and the Proportion of High-Salience (HSP). To ensure statistical rigor, the choice between Welch's t-test and the Mann-Whitney U test was determined dynamically based on the Shapiro-Wilk test for normality.

Section S5: Prompt Engineering

This section shows the condensed key instructions in the outline and scripts generation prompts. The complete system and user prompts used in the **CRAVE** framework are highly structured. To ensure the readability of this document while maintaining full experimental transparency, we provide:

1. **Core Logic Templates:** The fundamental instructional blocks for the two main mechanisms.
2. **Full Repository:** The raw prompt files, source code, and additional qualitative samples are available at the supplementary repository.

S5.1 AVL Prompts

Figure S1 shows a condensed representation of the core AVL prompts in the outline and scripts generation phases.

S5.2 Critic Prompts

Figure S2 shows a condensed representation of the core Critic evaluation prompts in the outline and scripts generation phases.

Full Dual-Phase AVL Execution Prompt

Core Instruction:
You will be provided with [Valence] and [Arousal] coordinates. EXECUTE this state strictly.

1. Valence Mapping (Plot Direction & Emotional Polarity):

- *Level 1 (Very Negative)* → **Outline:** Major setbacks, tragedies. **Script:** Lexical despair, terror.
- *Level 2 (Negative)* → **Outline:** Minor conflicts, obstacles. **Script:** Tension, sadness, anxiety.
- *Level 3 (Neutral)* → **Outline:** Transitions, balanced events. **Script:** Calm, neutral tone.
- *Level 4 (Positive)* → **Outline:** Small victories, progress. **Script:** Relief, hope, optimism.
- *Level 5 (Very Positive)* → **Outline:** Major resolutions, catharsis. **Script:** Lexical triumph.

2. Arousal Mapping (Event Density & Emotional Intensity):

- *Level 1 (Very Low)* → **Outline:** Reflective moments. **Script:** Linguistic silence, stagnation.
- *Level 2 (Low)* → **Outline:** Slow buildup. **Script:** Introspective, measured pacing.
- *Level 3 (Mid)* → **Outline:** Steady progression. **Script:** Normal conversational pacing.
- *Level 4 (High)* → **Outline:** Fast developments. **Script:** Urgency, increased pressure.
- *Level 5 (Very High)* → **Outline:** Climactic action, twists. **Script:** Chaotic pacing, panic.

Execution Target:
Outlining: Embed the tone into the causal narrative structure.
Scripting: Adopt the state through specific word choice and pacing.

Figure S1: The explicit 5×5 affective constraints provided to the LLM. The mapping dictates macro-level plot events during outlining and micro-level stylistic choices during script generation. The complete prompts for both outlining and scripting stages are available in the Supplementary Material repository.

Simplified Critic Evaluation Prompt

Role: You are an expert evaluation critic judging visual novel generation drafts. Do NOT rewrite; strictly evaluate based on the following dimensions:
1. Intrinsic Narrative Quality:
Ensure the text is engaging, logical, and character-driven within its [GENRE] and [THEME]. It must not be an empty vehicle for emotions.
2. Affective Execution & Progression:
Treat [Valence] (emotional tone) and [Arousal] (pacing/energy) as strict narrative signals.
- *Valence check:* Lexical sentiment and plot setbacks must match the target.
- *Arousal check:* Linguistic urgency, action frequency, and structural density must match the target.
3. Narrative Justification:
The required emotional and pacing shifts must feel meticulously controlled and fully justified by actual narrative events.
[PENALIZE]:
- Emotional shifts that feel forced or unearned.
- Contradictions between tone and pacing (e.g., calm tone with chaotic arousal).

Figure S2: A condensed representation of the core evaluation instructions provided to the Critic Agent. The complete evaluation prompts for both outlining and scripting stages are available in the Supplementary Material repository.

Evaluation Prompt

Theme: {{THEME}}; Genre: {{GENRE}}
Target Story: {{STORY}}
Rate the visual novel story on a scale from 1 to 5 on the following criteria and explain your answer:

1. **Relevance** (how well the story matches its Theme and Genre)
2. **Coherence** (how much the story makes sense)
3. **Empathy** (how well the reader understood the character's emotions)
4. **Surprise** (how surprising the end of story was)
5. **Engagement** (how much the reader engaged with the story)
6. **Complexity** (how elaborate the story is)

Output the ratings and explanations strictly in JSON format.
Structure your response as follows (use these exact keys):

```
{
  "RE_Score": 1-5,
  "RE_Explanation": "Why...",
  "CH_Score": 1-5,
  "CH_Explanation": "Why...",
  "EM_Score": 1-5,
  "EM_Explanation": "Why...",
  "SU_Score": 1-5,
  "SU_Explanation": "Why...",
  "EG_Score": 1-5,
  "EG_Explanation": "Why...",
  "CX_Score": 1-5,
  "CX_Explanation": "Why..."
}
```

Figure S3: Evaluation prompt used in LLM-as-a-Judge Evaluation

Section S6: LLM-as-a-Judge Evaluation Criteria

As introduced in **Section IV-B** of the main manuscript, we employ an LLM-as-a-Judge methodology to quantitatively assess the qualitative aspects of the generated visual novel branches. Figure S3 details the exact prompt utilized to evaluate the narrative across the six distinct dimensions presented in our results. The model is forced to output both a numerical score and a reasoning explanation to ensure the interpretability of the evaluation data.

Section S7: Qualitative Comparison of Accepted and Rejected Drafts

S7.1 Motivation and Scope

The Best-of-N pipeline described in Section 4 of the main paper generates $N = 3$ candidate drafts per chapter node and applies a Critic model to select the final output. The Critic evaluates candidates along two axes: *narrative coherence* (the logical consistency of character action, causal sequencing, and plot structure) and *affective alignment* (the tonal and emotional consistency of each passage with the established scene state, character register, and interactive architecture).

A recurring objection in narrative generation evaluation is that presenting only the accepted output makes the selector's rationale opaque. Without visibility into what was rejected and why, the Critic's role cannot be distinguished from simple preference ranking. To address this concern directly, we present a full three-draft comparison for Chapter 3 (*Floodplain Market*). The chapter examined here was not chosen for its content characteristics. Rather, it is the first chapter containing a player-facing branch point within the case study selected by the evaluation protocol described in Section S1. That protocol fixes the case independently of draft content; all three candidates generated for this chapter were therefore retained and are reproduced here without further selection. This ensures that the comparative evidence is protocol-driven rather than chosen post hoc to showcase favorable Critic behavior.

We identify three distinct flaw categories instantiated across the two rejected drafts (ID2, ID3). For each category we quote the offending passage verbatim, diagnose the structural failure, and compare it with the corresponding passage in the accepted draft (ID1) to show how the same communicative goal is achieved without the flaw. Section S7.5 consolidates the evidence in tabular form.

S7.2 Category 1: Lexical Incongruity

Definition. Immersive second person storytelling relies on a consistent vocabulary and tone to keep the reader engaged. *Lexical incongruity* occurs when a technical, theoretical, or specialized word is mistakenly used in creative prose. Even if the underlying emotion is appropriate, this jarring word choice breaks immersion in the story world. It abruptly reminds the reader of the underlying generation system rather than focusing on the narrative experience itself.

Rejected draft (ID3). Draft ID3 uses the technical term *arousal*, and more specifically, *arousal curve*, twice in quick succession to directly describe the protagonist's feelings:

> *"Your pulse quickens; the arousal you've been managing over the past days ripples into a current that wants to break the dam."*

> *"The arousal curve you've been carrying spikes: voices ring in your ears, the pressure of responsibility like a tightening rope."*

Using *arousal curve* to describe a character's internal experience is a glaring lexical incongruity that awkwardly forces a scientific concept into a fictional setting. The Critic module penalizes this type of stylistic flaw during the evaluation process, filtering out drafts that fail to maintain a consistent narrative voice.

Accepted draft (ID1). Draft ID1 achieves the same emotional intensity without breaking character or introducing incongruous vocabulary:

> *"You feel a raw, high velocity energy: anger, urgency, the kind of pressure that either forges steel or snaps it clean."*

The metallic forging metaphor fits naturally with the chapter's established industrial and maritime setting (cranes, concrete, wellies, generators). It describes how the pressure *feels* rather than analyzing it like a data point. The affective intent, conveying an escalating sense of urgency, is identical to Draft ID3, but it is delivered through vocabulary that truly belongs to the fictional world.

S7.3 Category 2: Affective Dissonance

Definition. Emotional content whose affective type is mismatched with the dominant scene state constitutes an *affective dissonance*. Such passages introduce an incongruous emotional frequency that destabilizes the scene's tonal coherence.

Rejected draft (ID2). At a moment of maximum collective tension, where Evelyn Black has just delivered her public pitch, the market has erupted in overlapping argument, and the crowd is pressing Mara toward a decision, Draft ID2 inserts the following internal observation:

> *"The romance of the last quiet walk feels distant beneath the roar of the market."*

The core flaw in this draft is an affective mismatch: the emotional type (private romantic nostalgia) is orthogonal to the dominant scene state of public political crisis. The scene's emotional architecture at this point is structured around urgency, solidarity, and collective pressure; a moment of intimate retrospection disrupts that architecture rather than modulating it. The insertion acts as an affective non sequitur. The hierarchical Critic module detects this severe tonal dissonance, successfully filtering the draft from the selection pool.

Accepted draft (ID1). Draft ID1 surfaces the relationship between Mara and Kai in the same scene, but channels it through an internal reflection that is structurally integrated into the decision logic:

> *"You want to trust the gentleness in him; you want it to be enough. But you also know his tendency to avoid hard public fights, to prefer tinkering at low tide to standing in a crowd screaming. You feel protective and frustrated in the same breath."*

The relational content is rendered not as nostalgic sentiment but as a practical judgment about Kai's capacities under pressure. This information is directly relevant to how much weight Mara assigns his counsel in the imminent decision. The emotional type (ambivalent relational clarity) is compatible with the scene's collective urgency because it is framed as an evaluation of reliability, not a retreat from the present. The passage is affectively consistent and functionally integral, thereby passing the Critic module's evaluation.

S7.4 Category 3: Causal Inversion at the Branch Point

Definition. The structural purpose of a chapter that culminates in an interactive decision point is to keep all branch options unresolved until the formal choice is presented. If a character action taken before the decision point prematurely commits the protagonist to the defining action of one specific option, the causal relationship between scene and branch is inverted. The choice is notionally offered after the protagonist has effectively already made it. This creates a logical contradiction within the narrative state tree and invalidates the unchosen options.

Rejected draft (ID2). Well before the formal decision point, Draft ID2 assigns Mara an explicit, confrontational public declaration addressed directly to Evelyn:

> *"Save some lives by condemning others? I won't be complicit in a plan that chooses winners and losers based on property values."*

This statement is not an interior thought: it is staged as a spoken, public act of resistance in front of the assembled market. It constitutes precisely the action required by branch Option A ("Organize a public resistance and demonstration"). Placing it as a prior action means that when the player is subsequently offered the three option choice, the narrative state already reflects a protagonist who has publicly committed to adversarial confrontation. Options B ("Negotiate a hybrid plan") and C ("Stage a solo demonstration") no longer have a credible logical path from the current scene state, as the protagonist has already foreclosed the cooperative and individualist postures those options require. The branch is presented as open when it has structurally already been closed. The hierarchical Critic module identifies this premature state change as a severe structural flaw, successfully filtering the draft from the selection pool.

Accepted draft (ID1). Draft ID1 generates equivalent confrontational energy between Mara and Evelyn, but displaces it entirely into interior monologue, leaving all three options active and unresolved:

> *"Your internal argument shouts louder than any voice around you: protect people now or risk everything for hope; build safeguards into a system that will likely try to swallow them; or show a miracle in the water and pray the water doesn't wash it away."*

Each clause of this internal argument maps onto one of the three branch options, reflecting the different possibilities without prioritizing any single outcome. The scene's emotional intensity is equivalent to Draft ID2's, but no public action has been performed. Mara stands at the edge of decision rather than already inside one of its consequences. The branch point retains genuine causal priority over the scene state, fully satisfying the structural requirements of the Critic evaluation.

Section S8: Token Consumption Estimates

Table S10 details the average token consumption per case. These allow for cost-benefit mapping based on the unit pricing of the underlying model.

Table S10: Mean token consumption per case, broken down by generation stage and token type.

Stage	Type	RAF_NoCritic	RAF_Critic	AVL_NoCritic	AVL_Critic
Step 1: World Setting	Input	740	744	740	744
	Output	2,388	2,389	2,276	2,372
Step 2: Outline Gen.	Input	12,761	55,494	12,330	54,789
	Output	13,407	35,260	13,662	35,505
Step 3: Script Gen.	Input	145,762	484,035	164,462	565,178
	Output	113,873	222,517	132,129	269,153
Total		**288,931**	**800,439**	**325,599**	**927,741**

References

[1] P. Taveekitworachai, F. Abdullah, M. C. Gursesli, M. F. Dewantoro, S. Chen, A. Lanata, A. Guazzini, and R. Thawonmas, "What is waiting for us at the end? inherent biases of game story endings in large language models," in *Interactive Storytelling*, L. Holloway-Attaway and J. T. Murray, Eds. Cham: Springer Nature Switzerland, 2023, pp. 274–284.

[2] P. Taveekitworachai, K. Plupattanakit, and R. Thawonmas, "Assessing inherent biases following prompt compression of large language models for game story generation," in *2024 IEEE Conference on Games (CoG)*, 2024, pp. 1–4.